Make it light up

IoT · Level 2 · Lesson 1



Today you will

- Build a complete circuit with a battery, wires and a bulb or LED.
- Sort materials into conductors and insulators by testing them.
- Explain why an LED needs a resistor and why a short circuit is dangerous.

Loop, LED, resistor

- A circuit is a closed loop
- Conductors let current through
- An LED's long leg goes to +
- An LED needs a resistor

Conductor hunt

1. Make a tester with a gap
2. Touch the free ends: check it lights
3. Predict ten things
4. Test each one and record it
5. Sort them and explain

Batteries only, never a wall socket. Ask an adult to cut or strip wires. Never join the battery's + and - with a bare wire or foil: that's a short circuit. Keep button batteries away from small children.

Let’s discuss

1. Which result surprised you, and why?
2. Why does an LED need a resistor when a torch bulb doesn't?
3. Why is a wire copper inside plastic, and not plastic inside copper?

Check yourself

- Name one conductor, one insulator, and one safety rule for circuits.